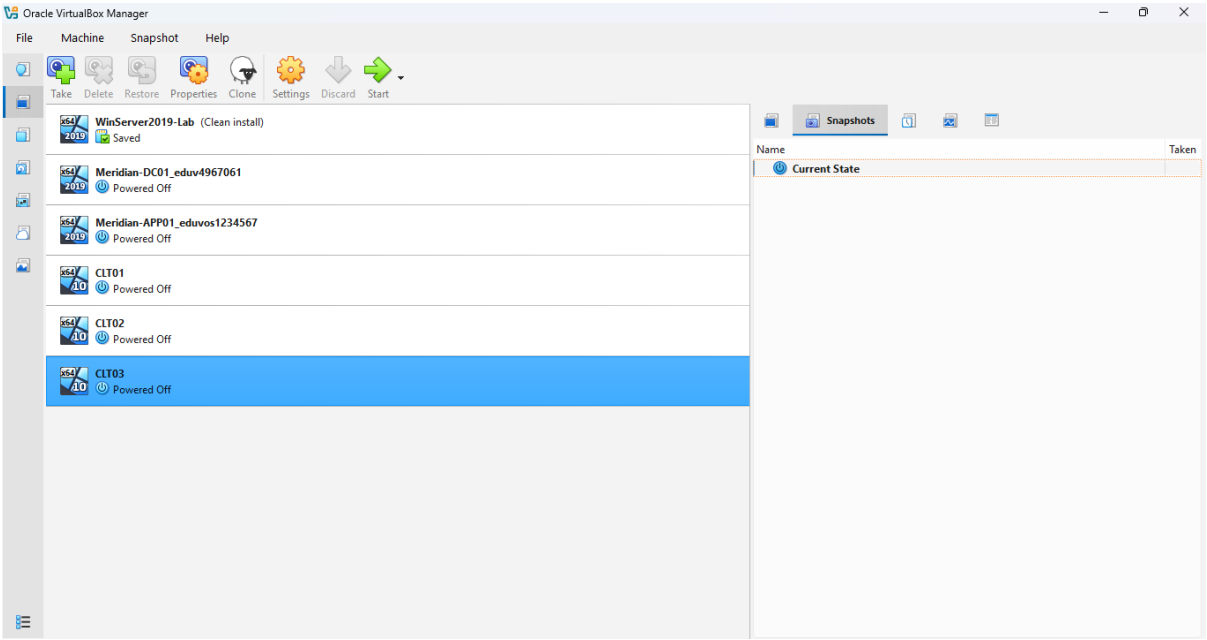
DELIVERABLE 1

1.1a)



The two window servers are:

- Meridian-APP01_1234567
- Meridian-DC01_eduv4967061

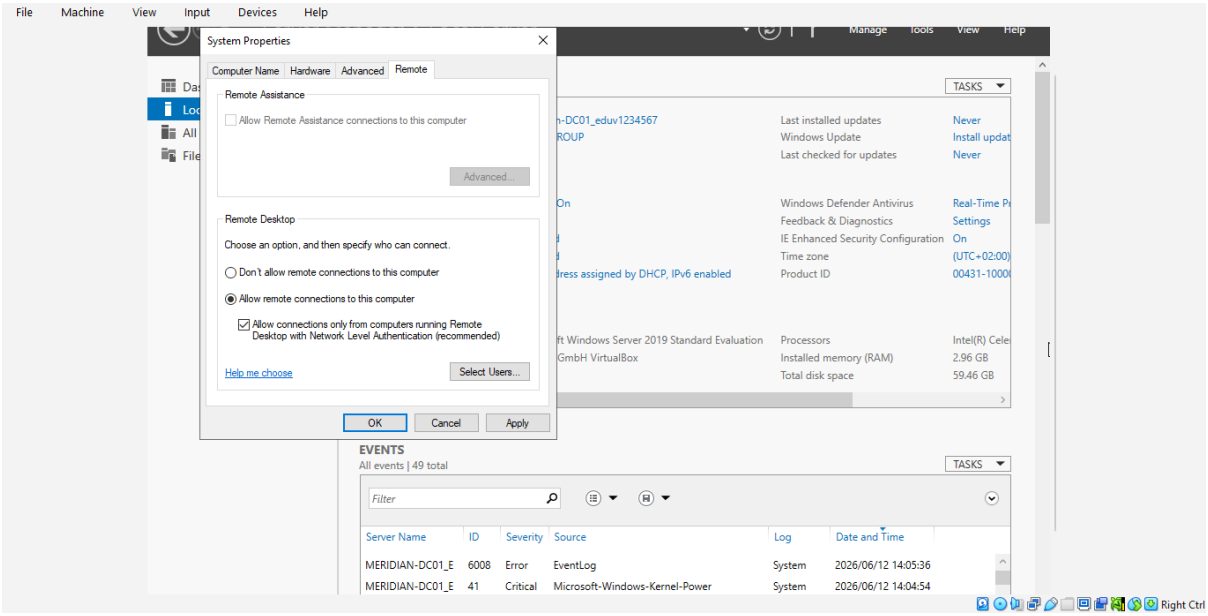
The three client machines are:

- CLT01
- CLT02
- CLT03

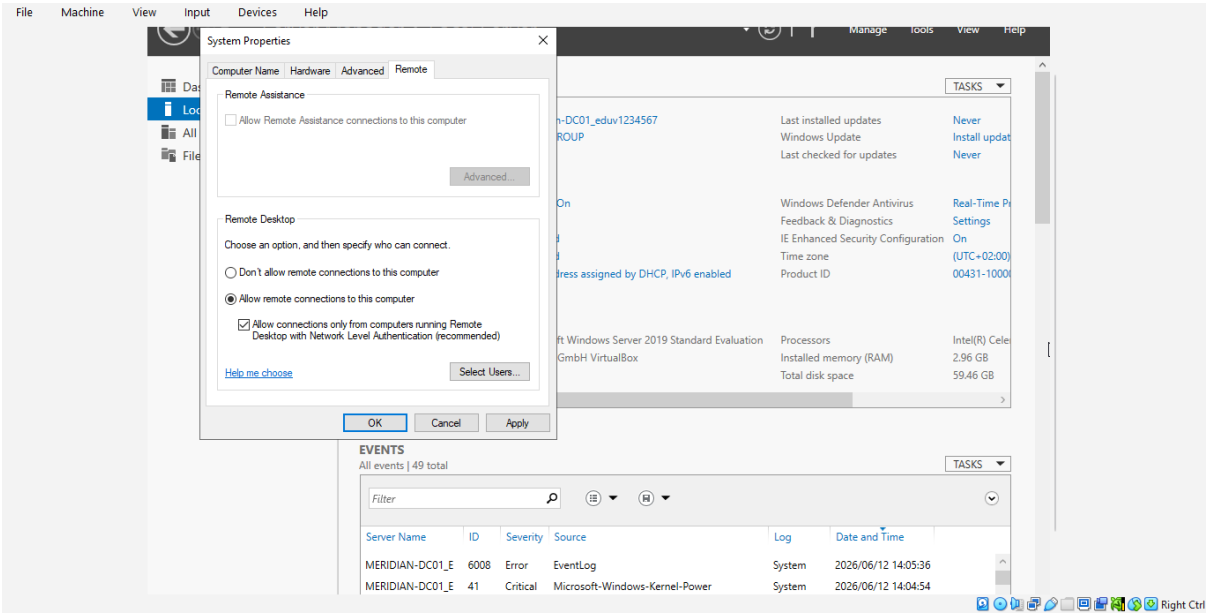
> ➢ ***Small disclaimer, I changed Meridian-DCO1_eduvos4967061 to Meridian-DC01_eduvos1234567 so don't be confused by that change. Its still my work***

b)

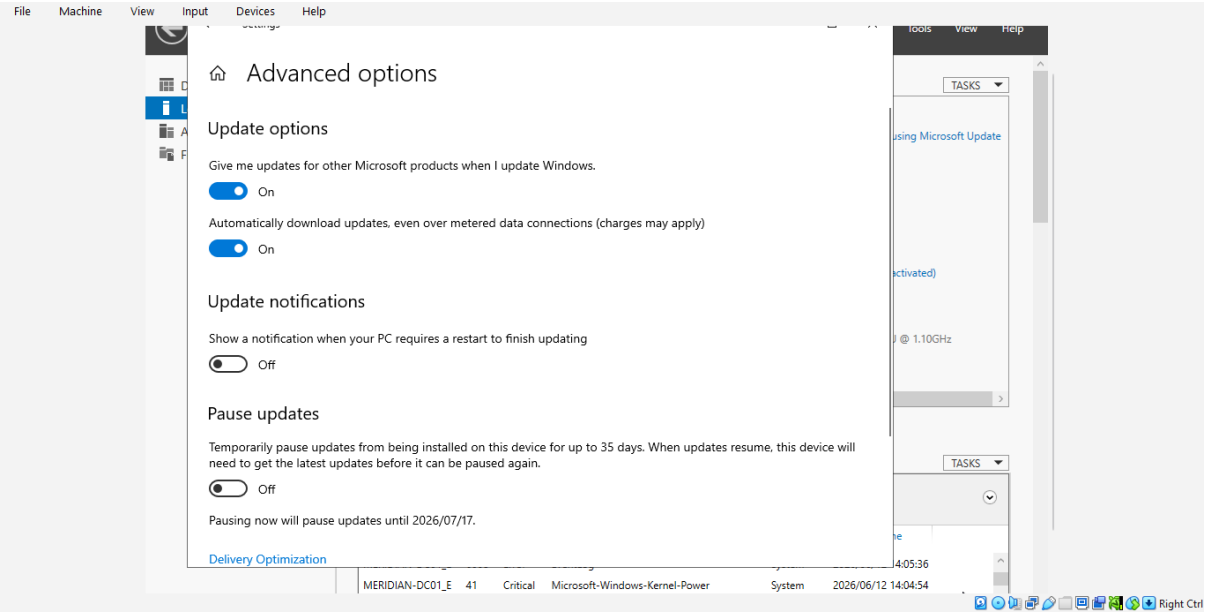
IP address:



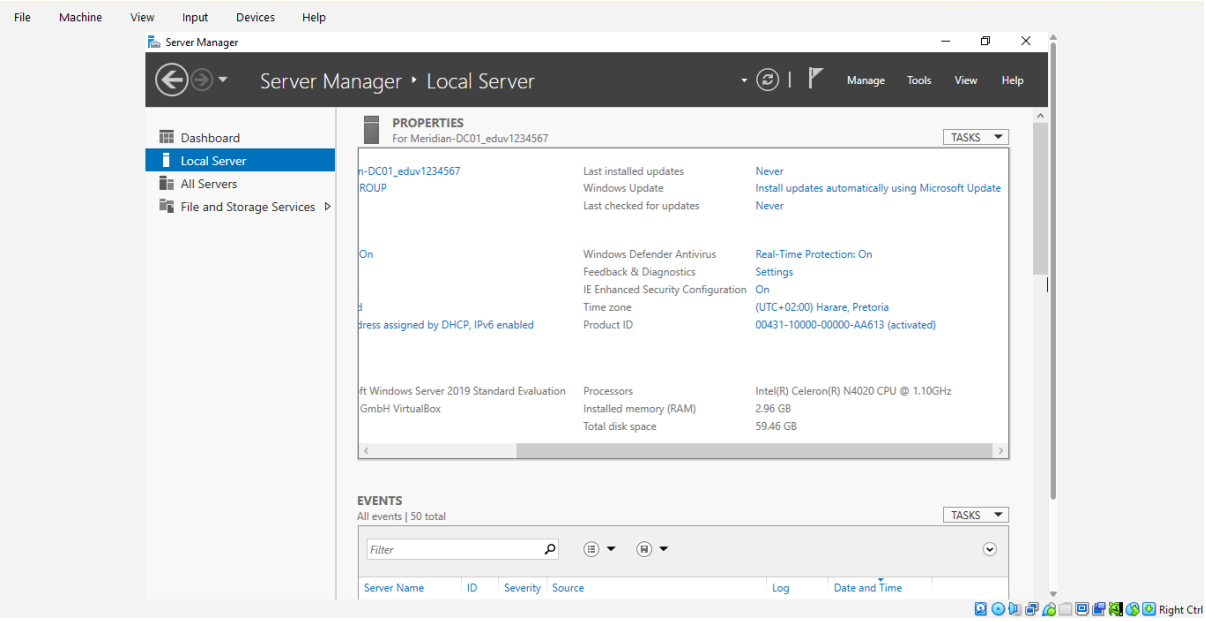
Remote desk:



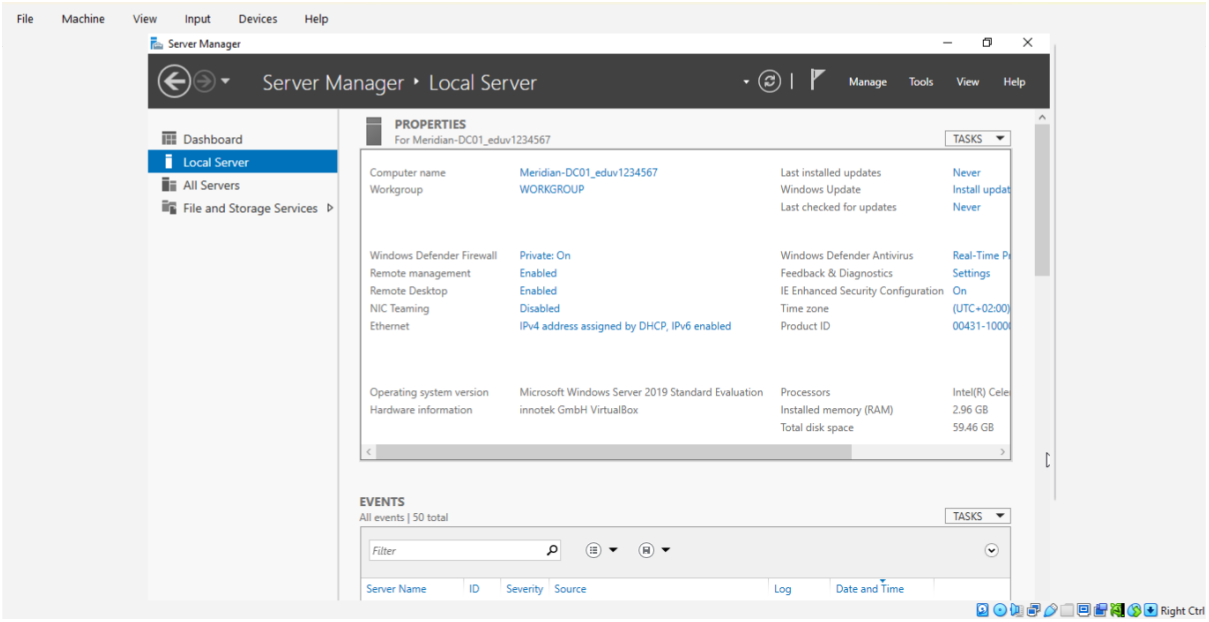
Automatic updates:



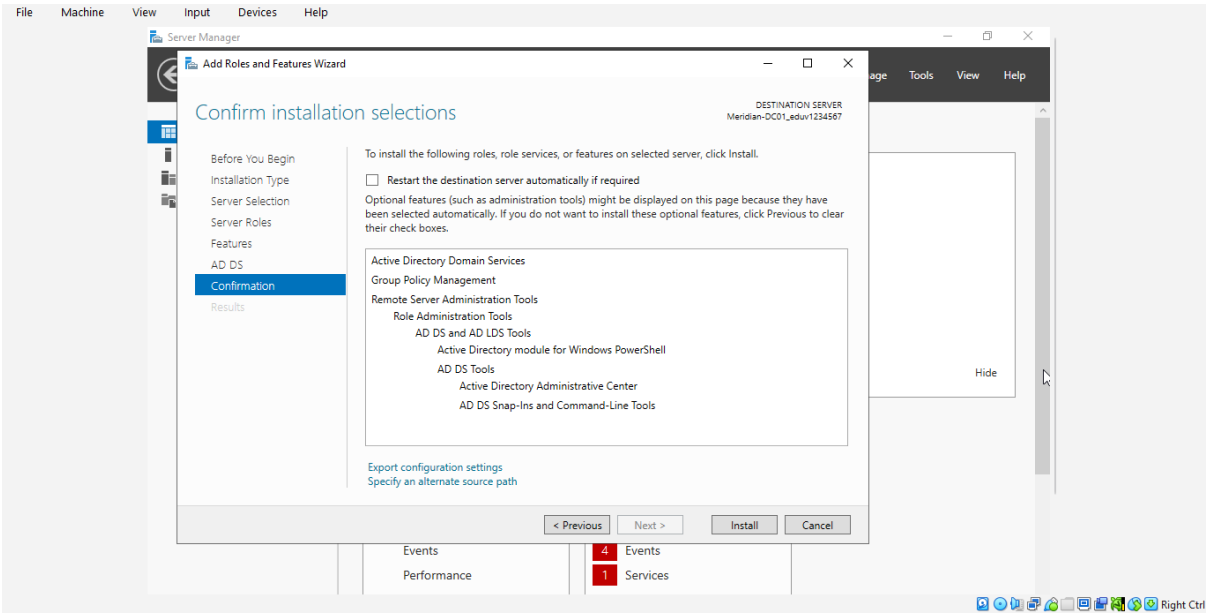
Time zone:

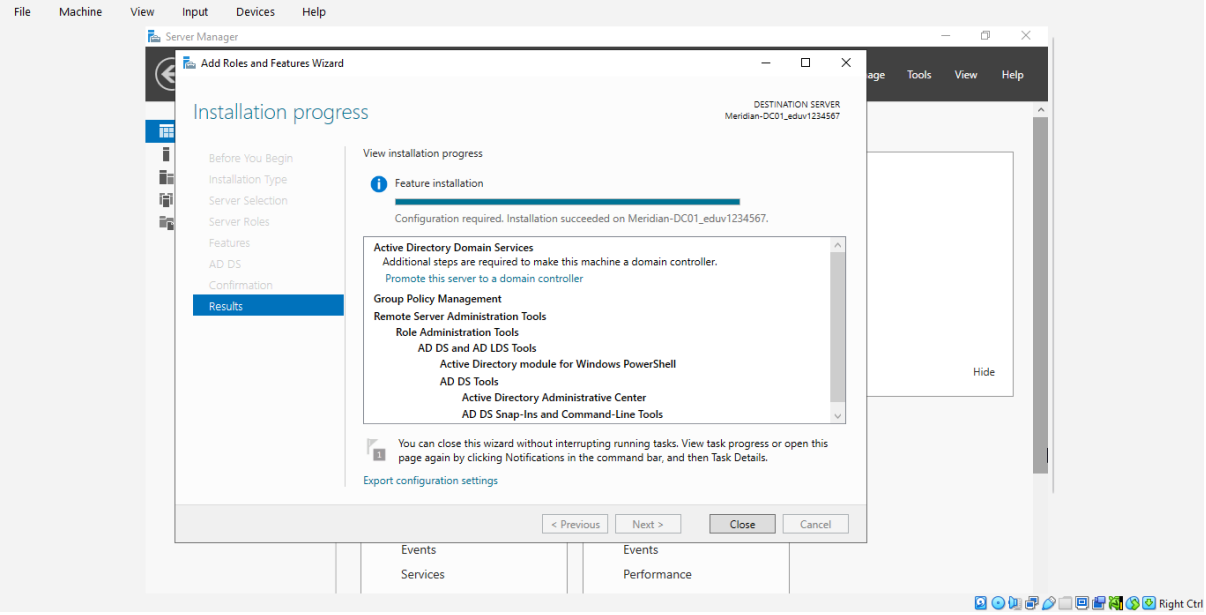


Hostname:


File Machine View Input Devices Help
Server Manager
Server Manager • Local Server
Manage Tools View Help
Dashboard
Local Server
All Servers
File and Storage Services
PROPERTIES
For Meridian-DC01_eduv1234567
TASKS
Computer name Meridian-DC01_eduv1234567
Workgroup WORKGROUP
Last installed updates Never
Windows Update Install updat
Last checked for updates Never
Windows Defender Firewall Private: On
Remote management Enabled
Remote Desktop Enabled
NIC Teaming Disabled
Ethernet IPv4 address assigned by DHCP, IPv6 enabled
Windows Defender Antivirus Real-Time P
Feedback & Diagnostics Settings
IE Enhanced Security Configuration On
Time zone (UTC+02:00)
Product ID 00431-1000
Operating system version Microsoft Windows Server 2019 Standard Evaluation
Hardware information innotek GmbH VirtualBox
Processors Intel(R) Cele
Installed memory (RAM) 2.96 GB
Total disk space 59.46 GB
EVENTS
All events | 50 total
TASKS
Filter
Server Name ID Severity Source Log Date and Time
Right Ctrl

c)

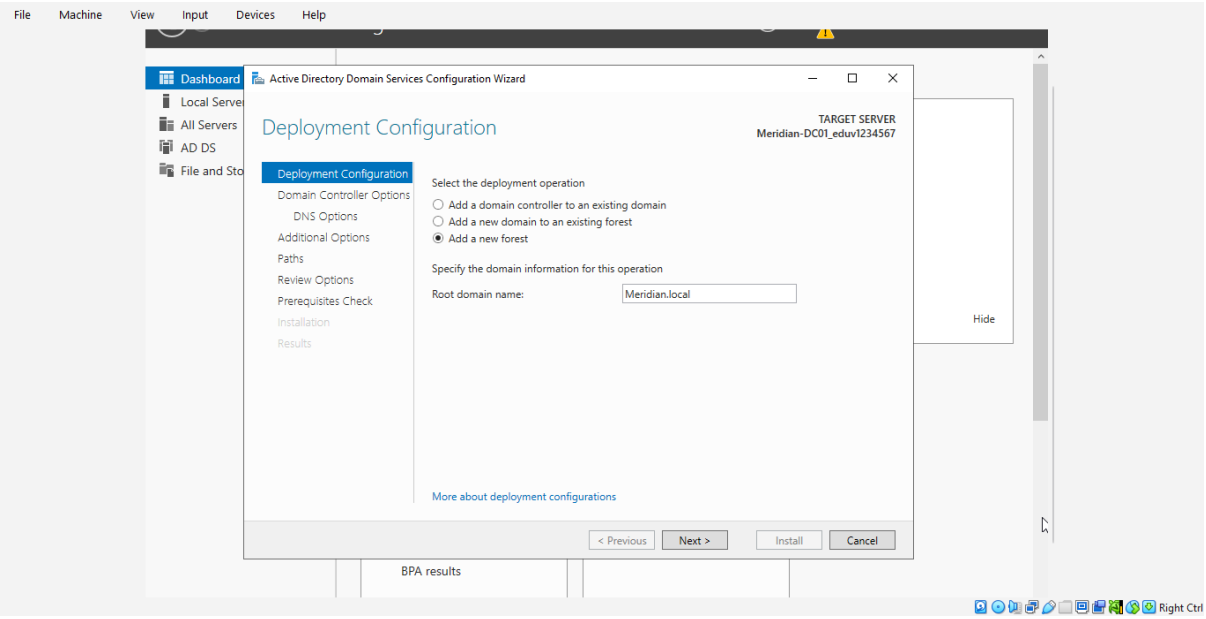

File Machine View Input Devices Help
Server Manager
Add Roles and Features Wizard
Confirm installation selections
DESTINATION SERVER
Meridian-DC01_eduv1234567
Before You Begin
Installation Type
Server Selection
Server Roles
Features
AD DS
Confirmation
Results
To install the following roles, role services, or features on selected server, click Install.
Restart the destination server automatically if required
Optional features (such as administration tools) might be displayed on this page because they have been selected automatically. If you do not want to install these optional features, click Previous to clear their check boxes.
Active Directory Domain Services
Group Policy Management
Remote Server Administration Tools
Role Administration Tools
AD DS and AD LDS Tools
Active Directory module for Windows PowerShell
AD DS Tools
Active Directory Administrative Center
AD DS Snap-Ins and Command-Line Tools
Export configuration settings
Specify an alternate source path
< Previous Next > Install Cancel
Hide
Events
Performance
4 Events
1 Services
Right Ctrl

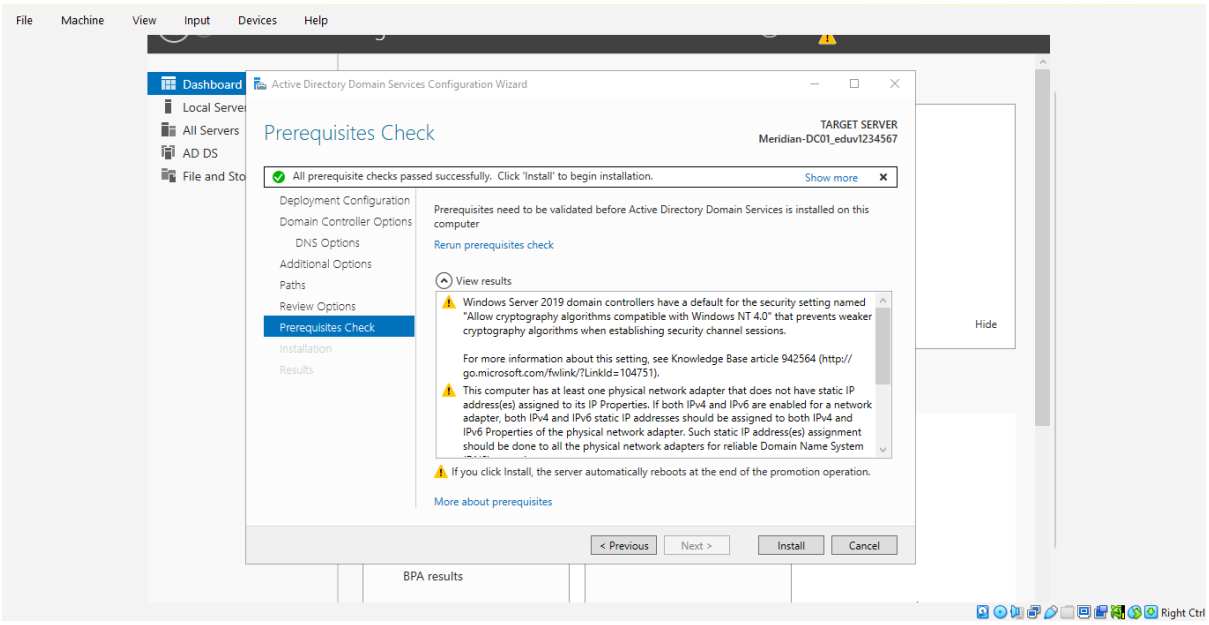


d) The reason why it can be beneficial is:

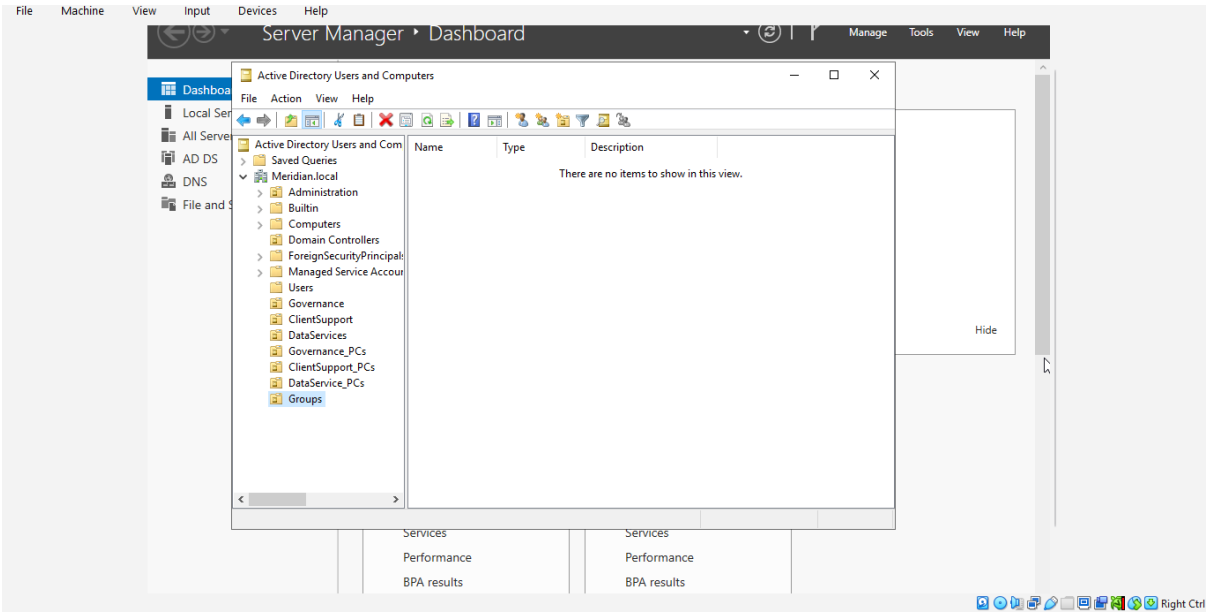
- It provides security and stability. The domain controller handles the authentication for the entire network. This keeps it dedicated of the web/file server not being compromised. It does not affect identity management and domain functionallity.
- It provides scalability. In a pilot that is intended for later expansion, it seperates roles which allow each other to be upgraded, cloned or replaced. Additional application servers can be added without touching the core infrastructure server which supports growth across multiple offices .

1.2a)

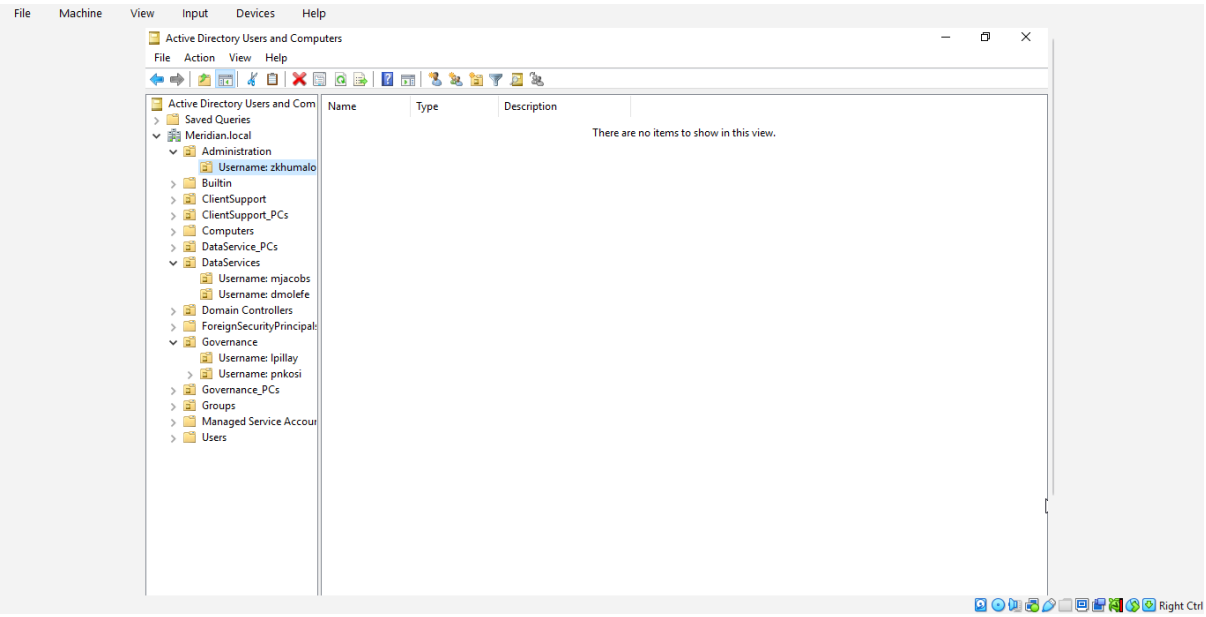



File Machine View Input Devices Help
Dashboard
Local Server
All Servers
AD DS
File and Sto
Active Directory Domain Services Configuration Wizard
Prerequisites Check
TARGET SERVER
Meridian-DC01_eduv1234567
All prerequisite checks passed successfully. Click 'Install' to begin installation.
Show more
Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
Prerequisites need to be validated before Active Directory Domain Services is installed on this computer
Rerun prerequisites check
View results
Windows Server 2019 domain controllers have a default for the security setting named "Allow cryptography algorithms compatible with Windows NT 4.0" that prevents weaker cryptography algorithms when establishing security channel sessions.
For more information about this setting, see Knowledge Base article 942564 (http://go.microsoft.com/fwlink/?LinkId=104751).
This computer has at least one physical network adapter that does not have static IP address(es) assigned to its IP Properties. If both IPv4 and IPv6 are enabled for a network adapter, both IPv4 and IPv6 static IP addresses should be assigned to both IPv4 and IPv6 Properties of the physical network adapter. Such static IP address(es) assignment should be done to all the physical network adapters for reliable Domain Name System
If you click Install, the server automatically reboots at the end of the promotion operation.
More about prerequisites
< Previous
Next >
Install
Cancel
Hide
BPA results
Right Ctrl

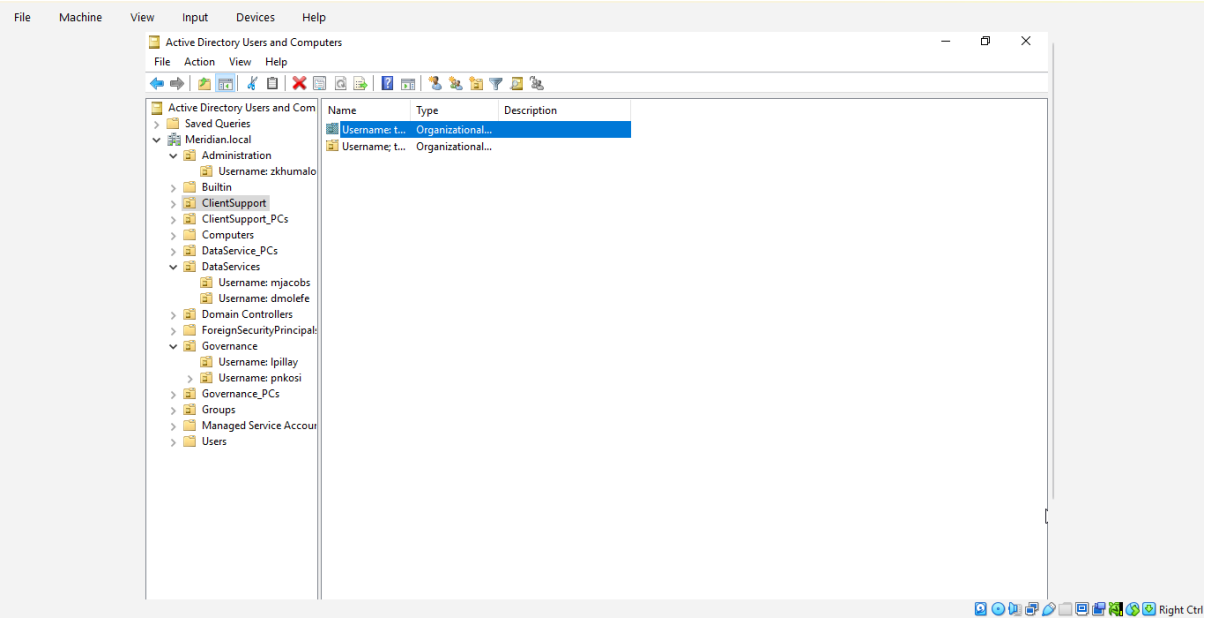
b)


File Machine View Input Devices Help
Server Manager • Dashboard
Manage Tools View Help
Active Directory Users and Computers
File Action View Help
Active Directory Users and Com
Saved Queries
Meridian.local
Administration
Builtin
Computers
Domain Controllers
ForeignSecurityPrincipals
Managed Service Accour
Users
Governance
ClientSupport
DataServices
Governance_PCs
ClientSupport_PCs
DataService_PCs
Groups
Name
Type
Description
There are no items to show in this view.
Hide
Performance
BPA results
Right Ctrl

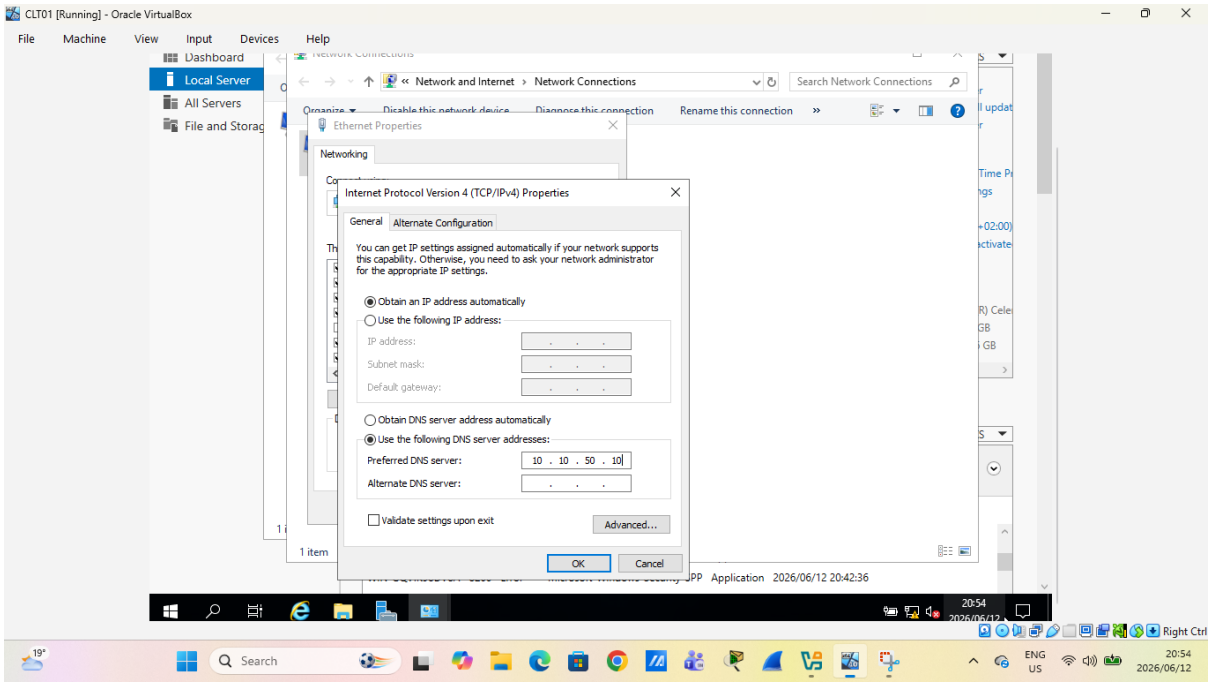
c)

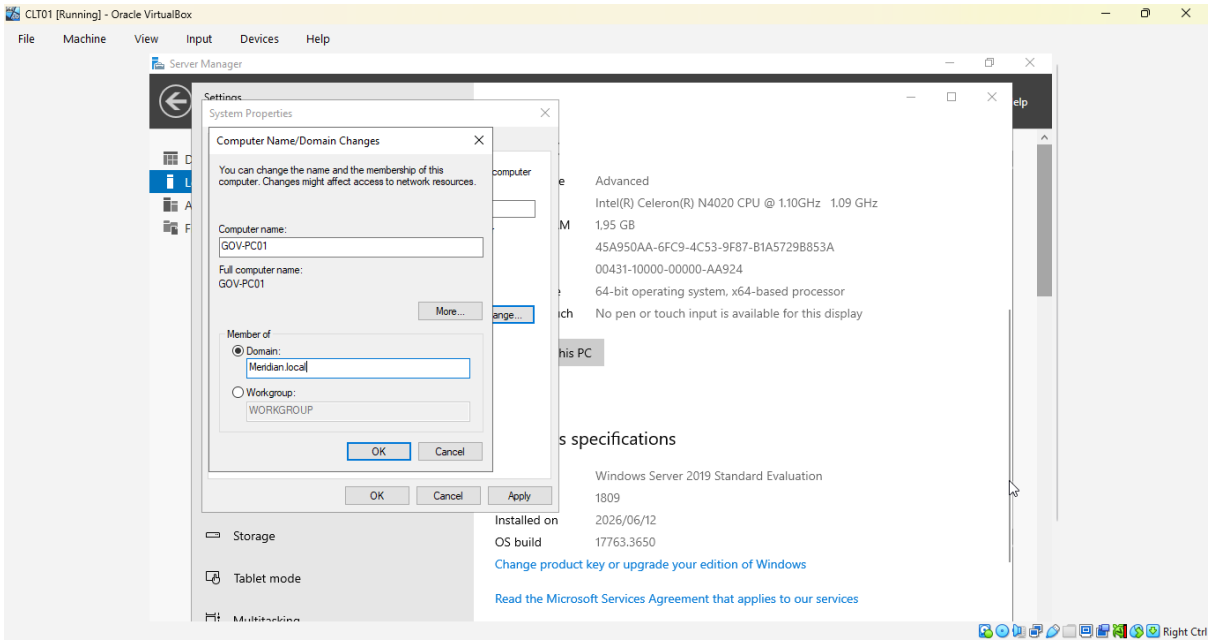


Client support staff account



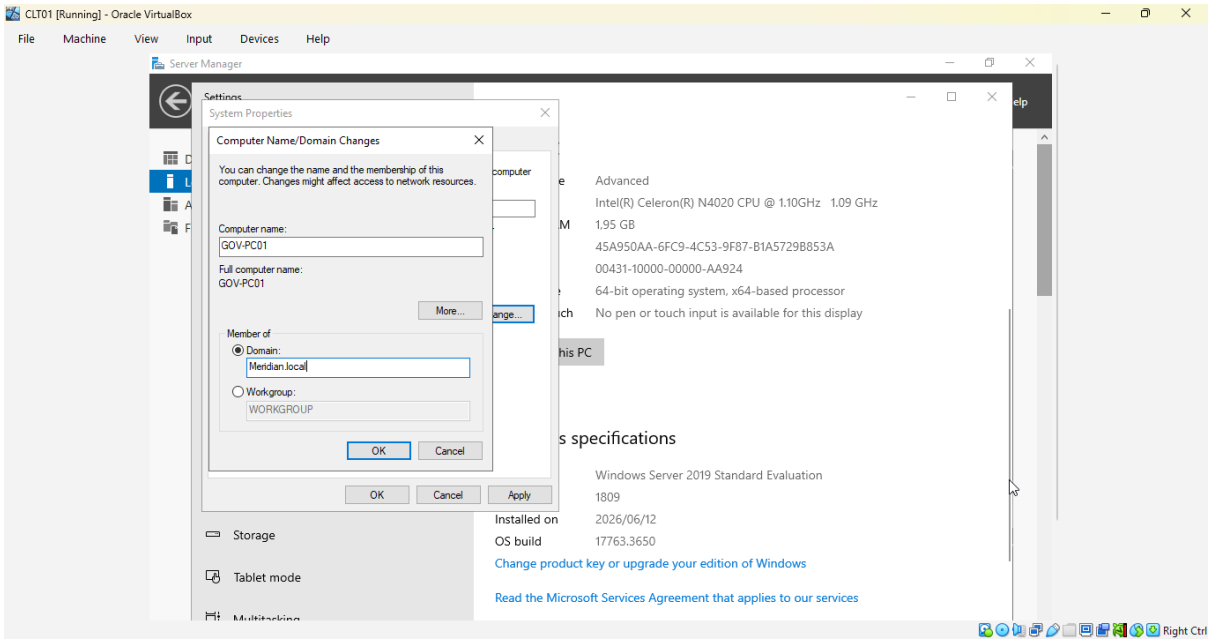
d)

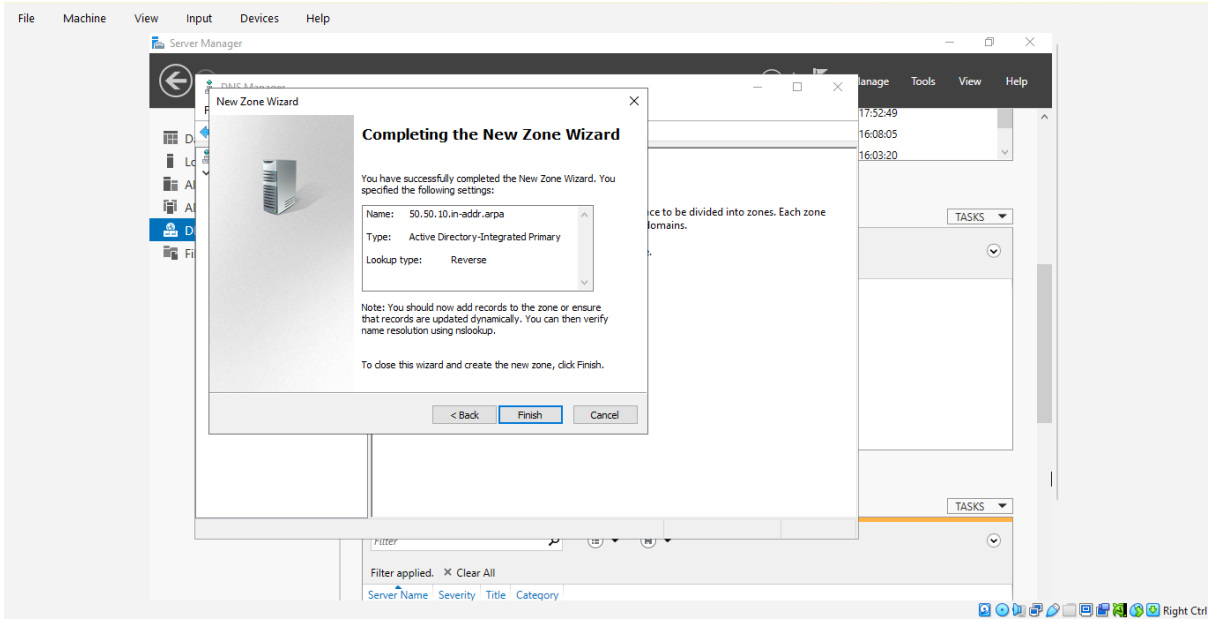

CLT01 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Dashboard
Local Server
All Servers
File and Storag
Network and Internet › Network Connections
Search Network Connections
Disable this network device
Diagnose this connection
Rename this connection
Ethernet Properties
Networking
Internet Protocol Version 4 (TCP/IPv4) Properties
General
Alternate Configuration
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address:
Subnet mask:
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server: 10 . 10 . 50 . 10
Alternate DNS server:
Validate settings upon exit
Advanced...
OK
Cancel
1 item
Application 2026/06/12 20:42:36
20:54
2026/06/12
Right Ctrl
Search
ENG
US
20:54
2026/06/12

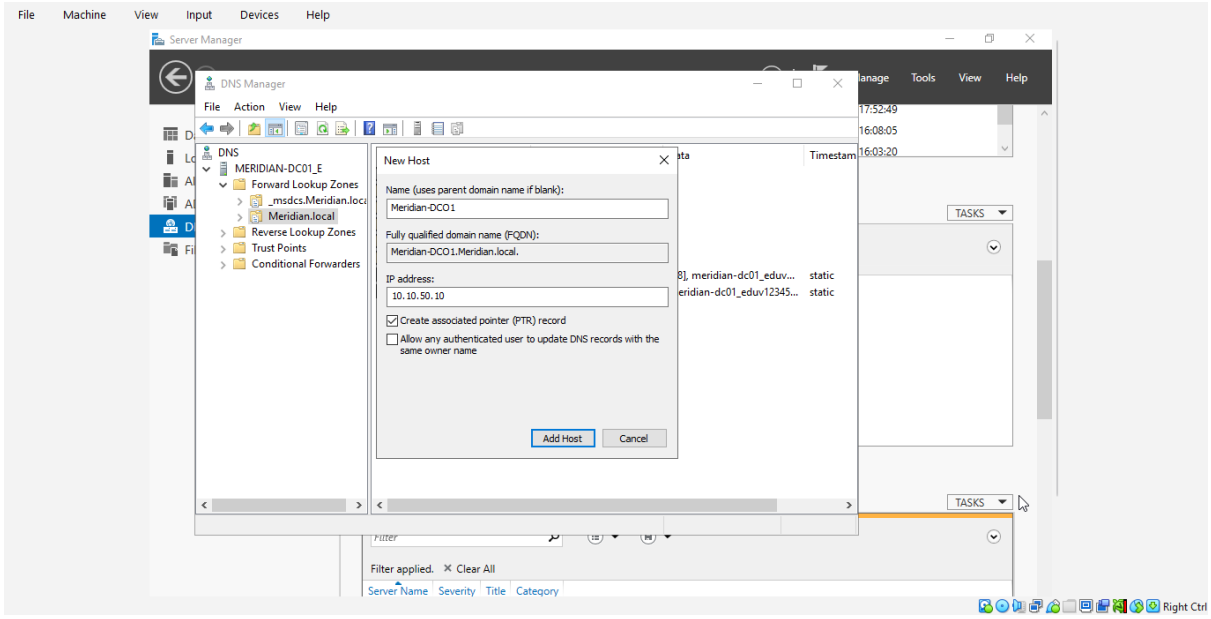

CLT01 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Server Manager
System Properties
Computer Name/Domain Changes
You can change the name and the membership of this computer. Changes might affect access to network resources.
Computer name:
GOV-PC01
Full computer name:
GOV-PC01
More...
Member of
Domain:
Meridian.local
Workgroup:
WORKGROUP
OK
Cancel
OK
Cancel
Apply
Advanced
Intel(R) Celeron(R) N4020 CPU @ 1.10GHz 1.09 GHz
1.95 GB
45A950AA-6FC9-4C53-9F87-B1A5729B853A
00431-10000-00000-AA924
64-bit operating system, x64-based processor
No pen or touch input is available for this display
s specifications
Windows Server 2019 Standard Evaluation
1809
Installed on 2026/06/12
OS build 17763.3650
Change product key or upgrade your edition of Windows
Read the Microsoft Services Agreement that applies to our services
Storage
Tablet mode
Right Ctrl

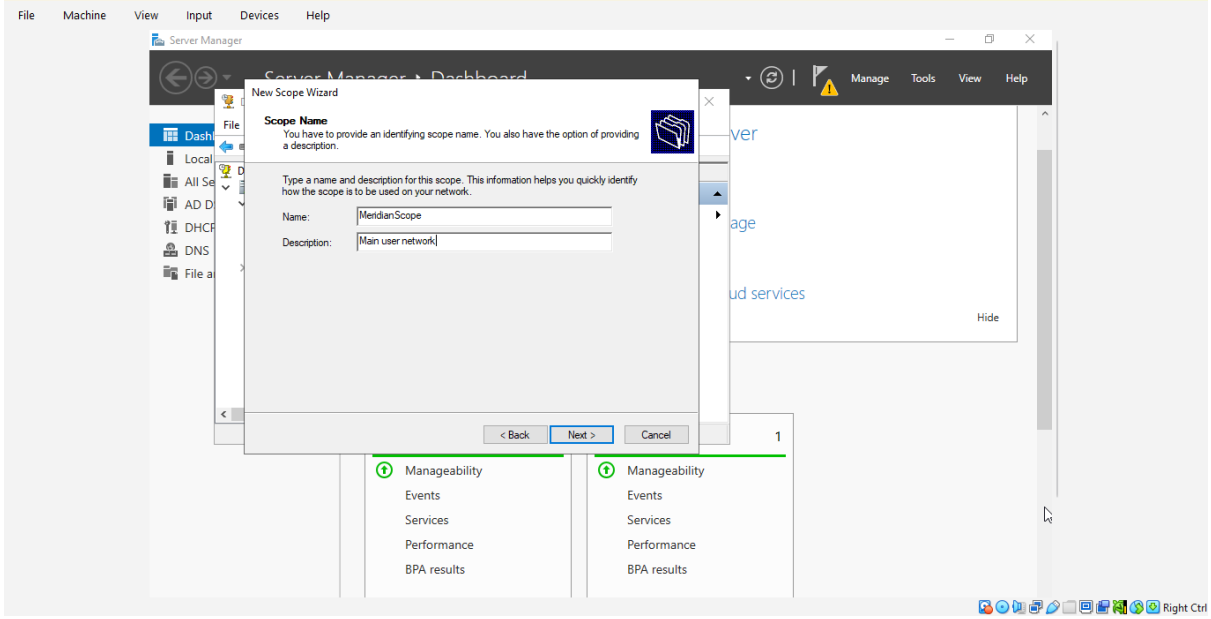
DELIVERABLE 2

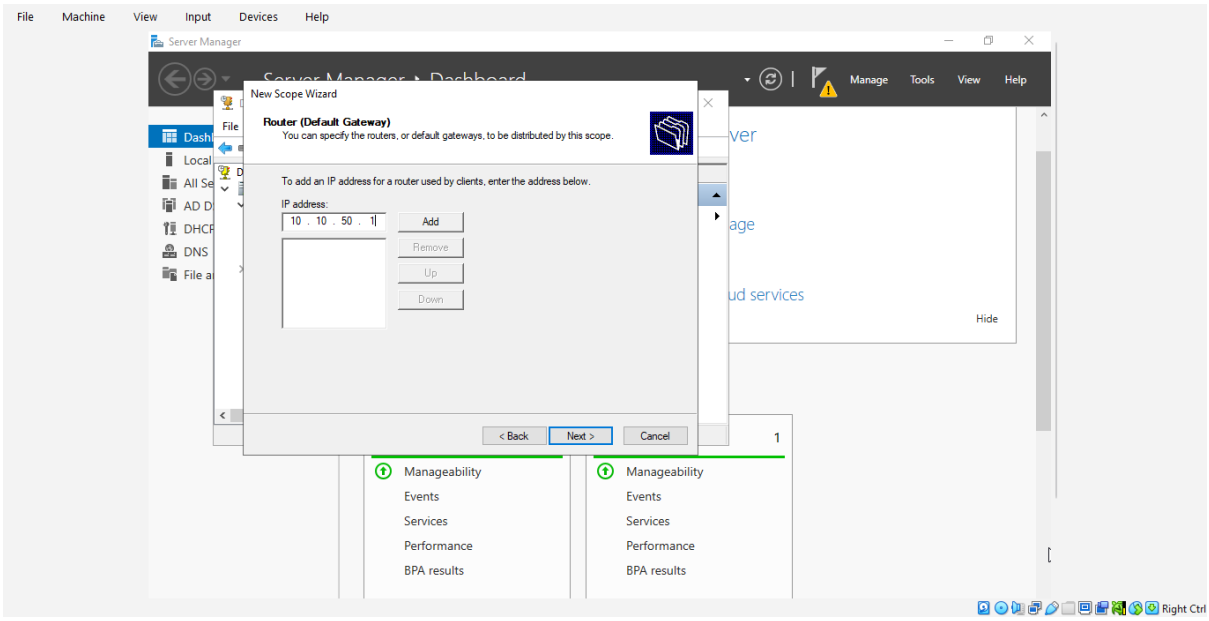
2.1a)

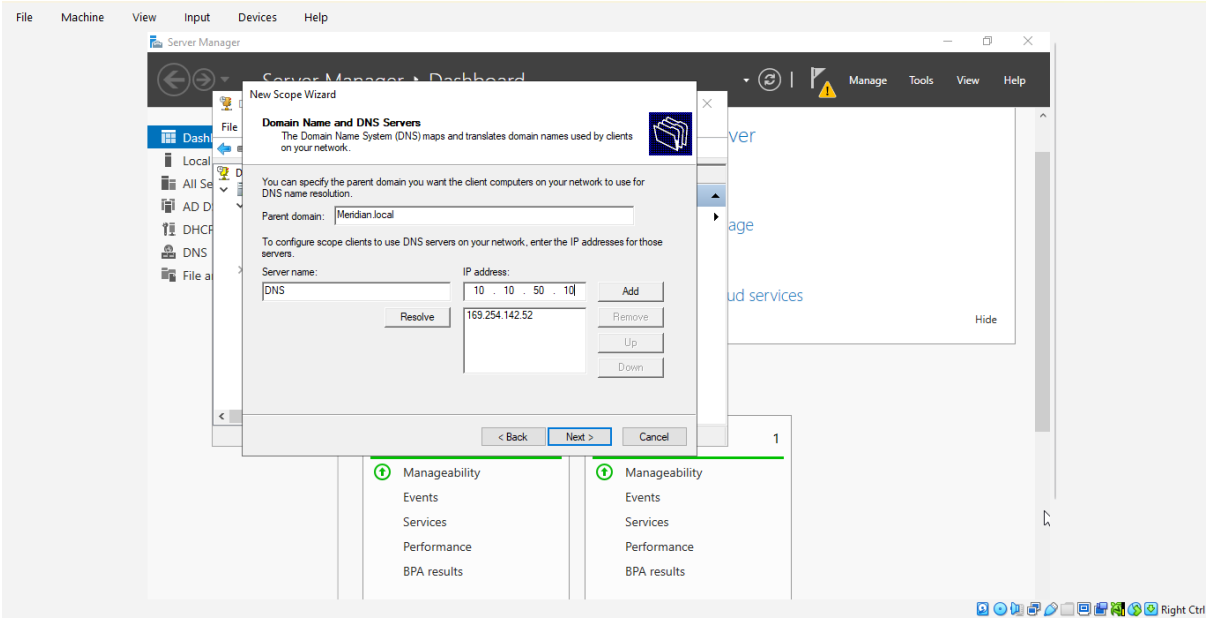


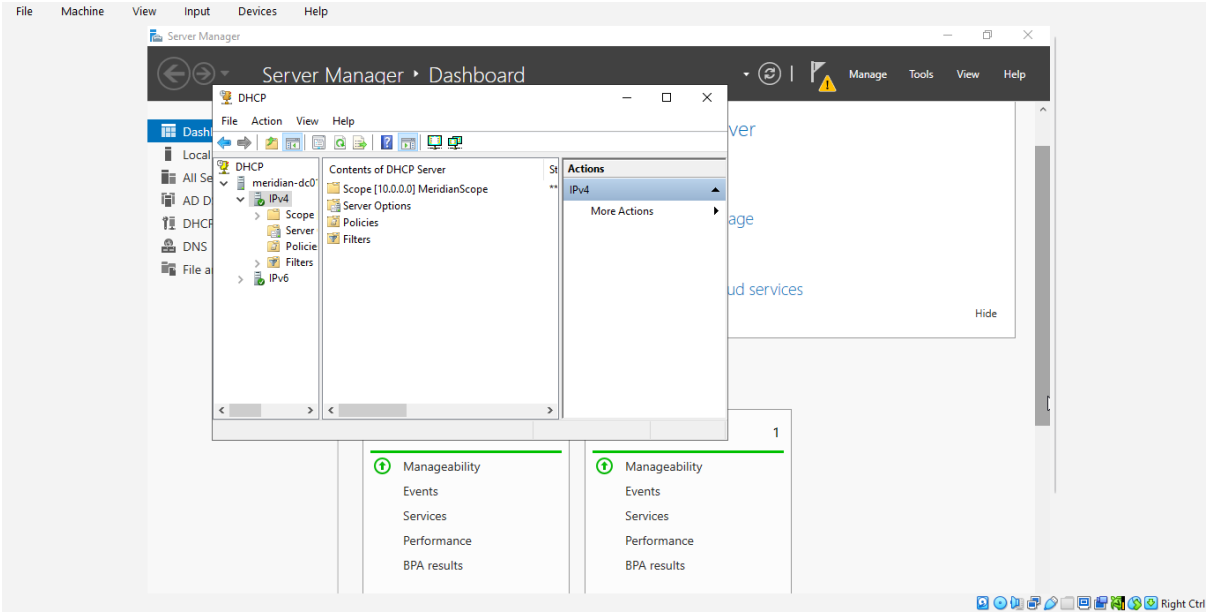



File Machine View Input Devices Help
Server Manager
DNS Manager
File Action View Help
DNS
MERIDIAN-DC01_E
Forward Lookup Zones
_msdcs.Meridian.loca
Meridian.local
Reverse Lookup Zones
Trust Points
Conditional Forwarders
New Host
Name (uses parent domain name if blank):
Meridian-DC01
Fully qualified domain name (FQDN):
Meridian-DC01.Meridian.local.
IP address:
10.10.50.10
Create associated pointer (PTR) record
Allow any authenticated user to update DNS records with the same owner name
Add Host
Cancel
Manage Tools View Help
TASKS
Filter applied. Clear All
Server Name Severity Title Category
Right Ctrl

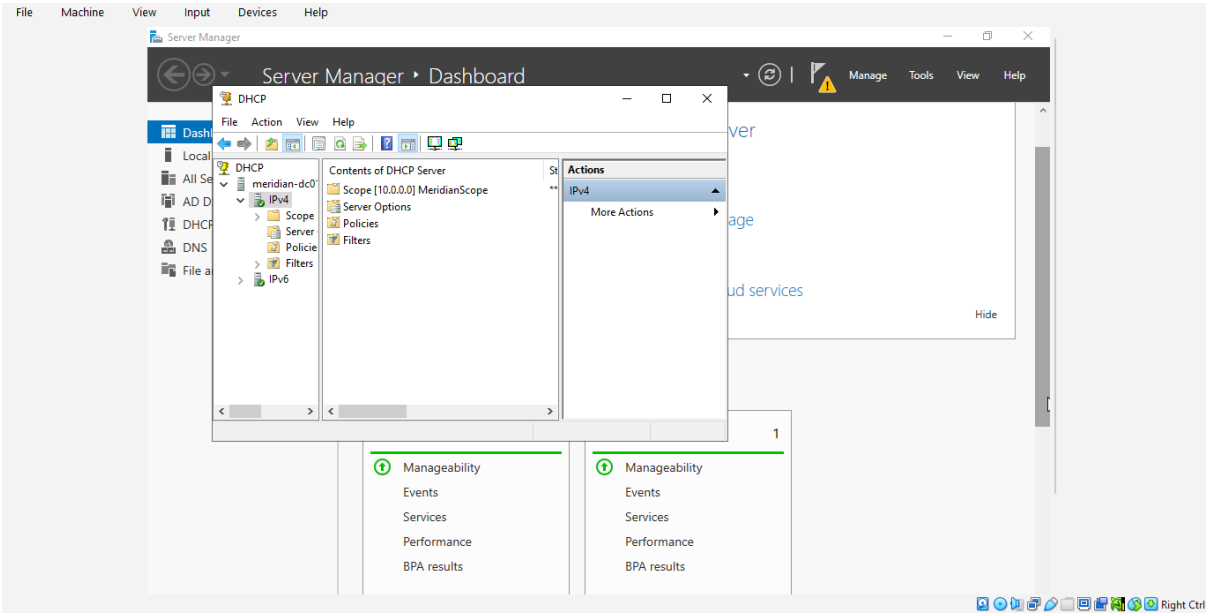

File Machine View Input Devices Help
Server Manager
Server Manager • Dashboard
Manage Tools View Help
New Scope Wizard
Scope Name
You have to provide an identifying scope name. You also have the option of providing a description.
Type a name and description for this scope. This information helps you quickly identify how the scope is to be used on your network.
Name:
MeridianScope
Description:
Main user network
< Back
Next >
Cancel
Hide
Manageability
Events
Services
Performance
BPA results
Right Ctrl


File Machine View Input Devices Help
Server Manager
Manage Tools View Help
New Scope Wizard
Router (Default Gateway)
You can specify the routers, or default gateways, to be distributed by this scope.
To add an IP address for a router used by clients, enter the address below.
IP address:
10 . 10 . 50 . 1
Add
Remove
Up
Down
< Back
Next >
Cancel
Manageability
Events
Services
Performance
BPA results
Hide
Right Ctrl

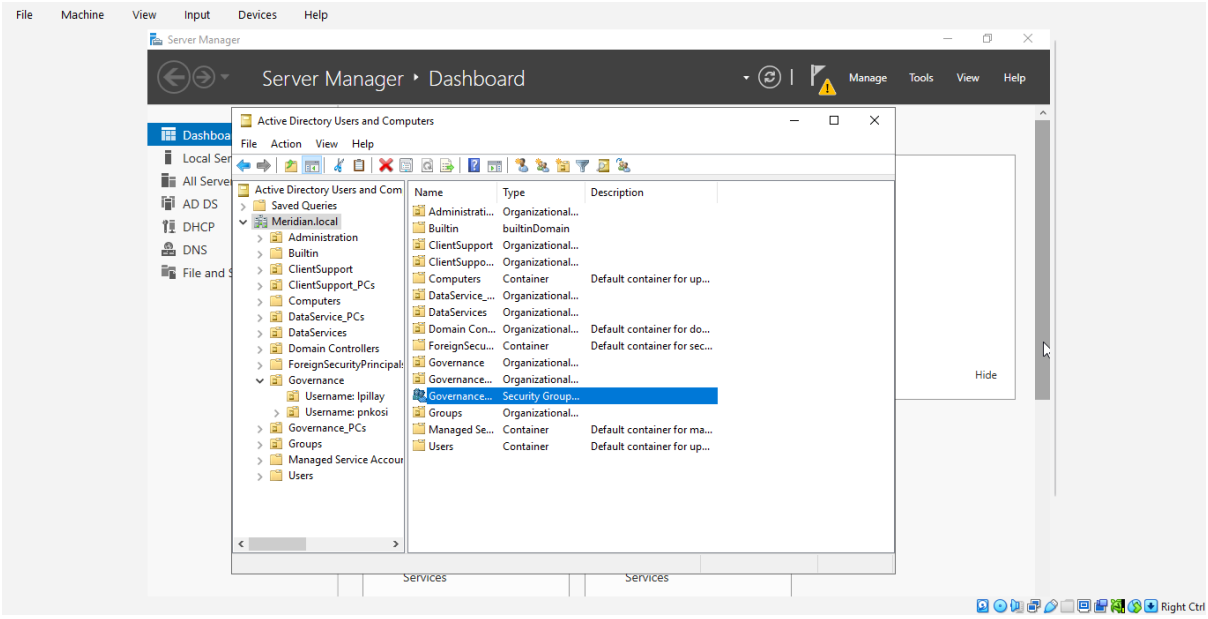

File Machine View Input Devices Help
Server Manager
Manage Tools View Help
New Scope Wizard
Domain Name and DNS Servers
The Domain Name System (DNS) maps and translates domain names used by clients on your network.
You can specify the parent domain you want the client computers on your network to use for DNS name resolution.
Parent domain:
Meridian.local
To configure scope clients to use DNS servers on your network, enter the IP addresses for those servers.
Server name:
DNS
Resolve
IP address:
10 . 10 . 50 . 10
Add
169.254.142.52
Remove
Up
Down
< Back
Next >
Cancel
Manageability
Events
Services
Performance
BPA results
Hide
Right Ctrl


File Machine View Input Devices Help
Server Manager
Server Manager ‣ Dashboard
Manage Tools View Help
DHCP
File Action View Help
DHCP
meridian-dc0
IPv4
Scope
Server
Policie
Filters
IPv6
Contents of DHCP Server
Scope [10.0.0.0] MeridianScope
Server Options
Policies
Filters
Actions
IPv4
More Actions
Hide
Manageability
Events
Services
Performance
BPA results
Right Ctrl

2.2)

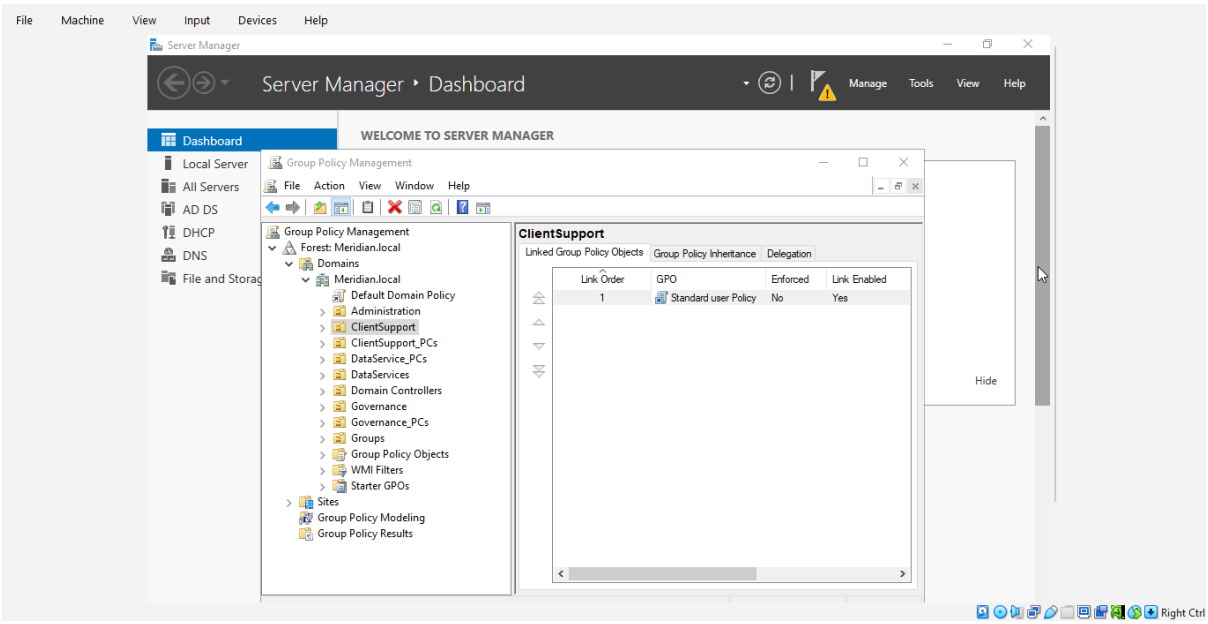

File Machine View Input Devices Help
Server Manager
Server Manager ‣ Dashboard
Manage Tools View Help
DHCP
File Action View Help
DHCP
meridian-dc0
IPv4
Scope
Server
Policie
Filters
IPv6
Contents of DHCP Server
Scope [10.0.0.0] MeridianScope
Server Options
Policies
Filters
Actions
IPv4
More Actions
Hide
Manageability
Events
Services
Performance
BPA results
Right Ctrl

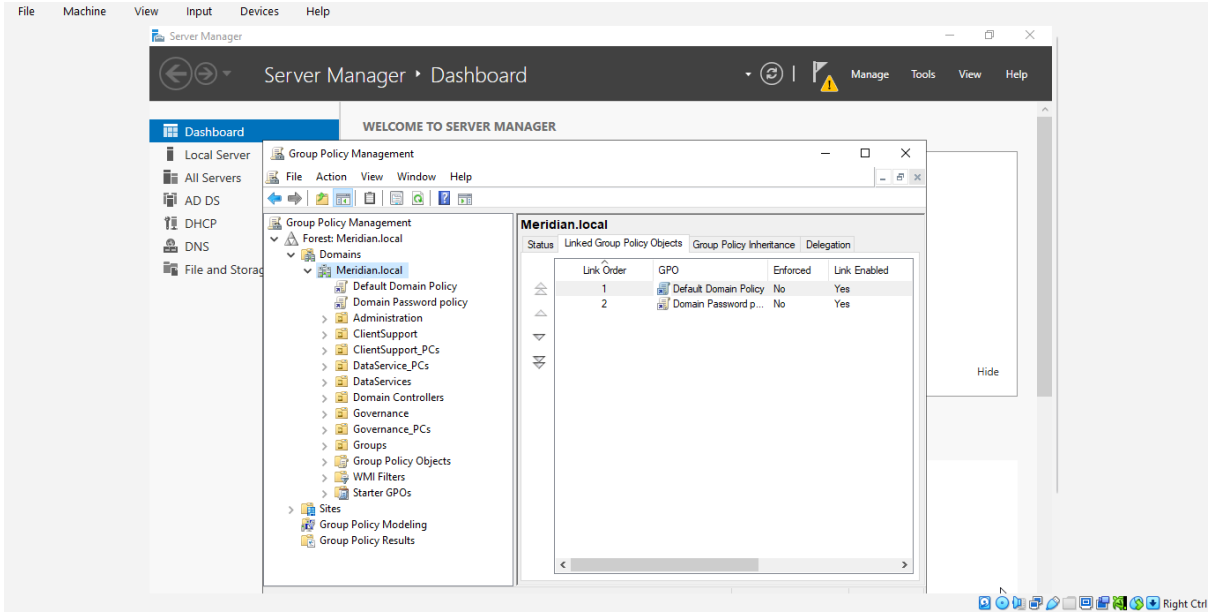
2.3)

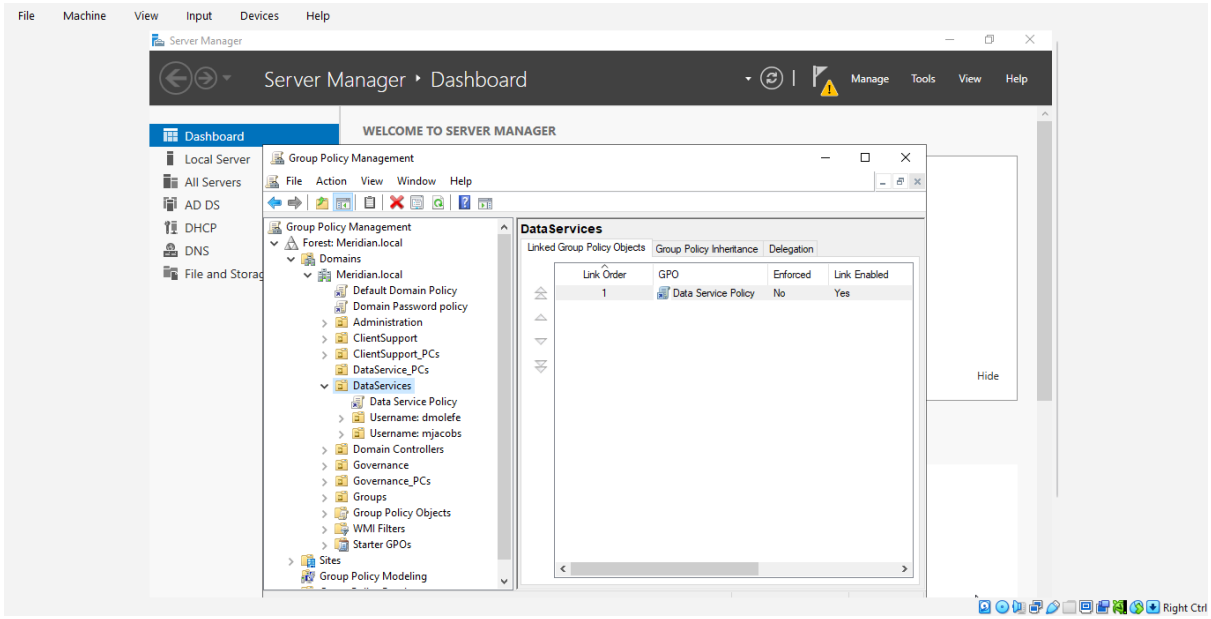


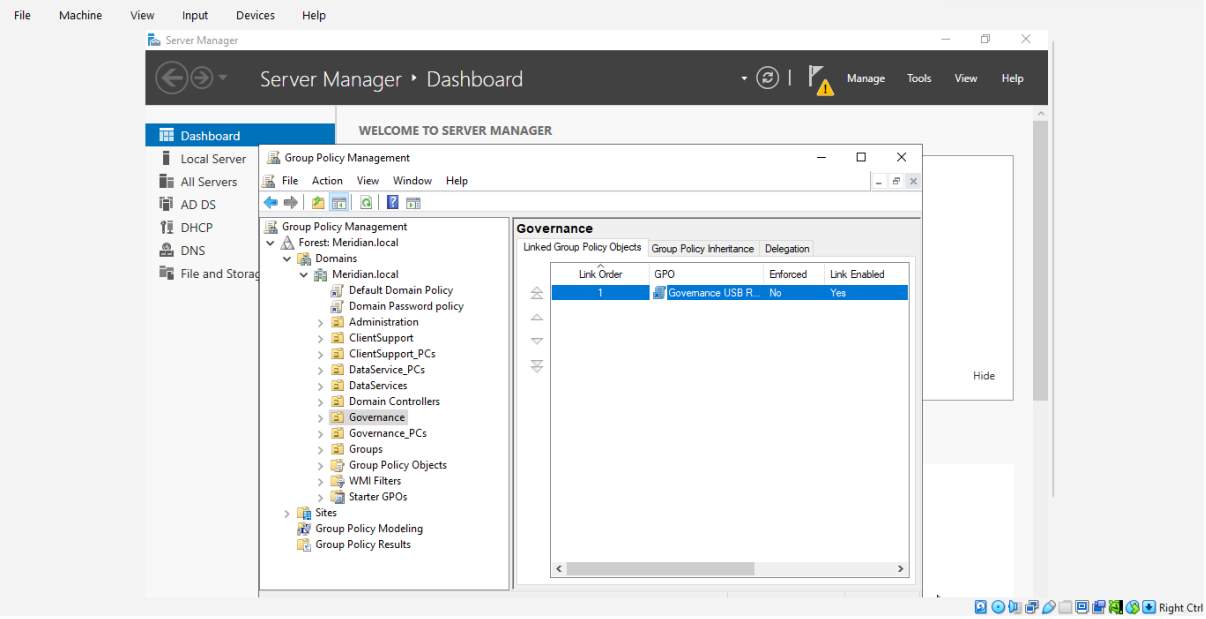
DELIVERABLE 3

3.1a)

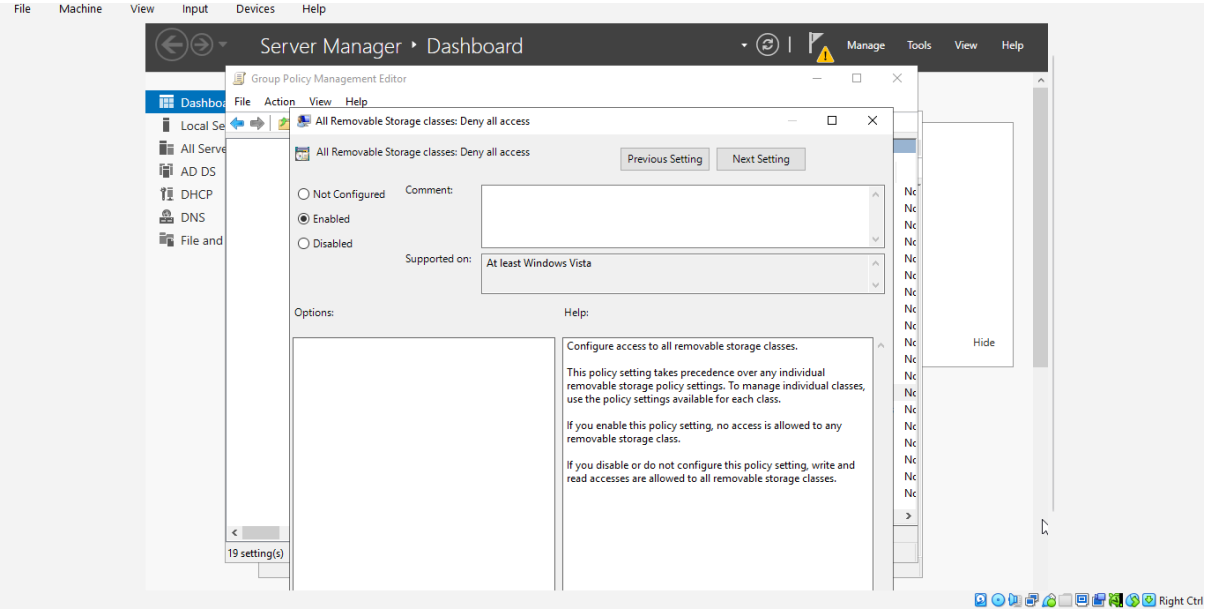



File Machine View Input Devices Help
Server Manager
Server Manager ‣ Dashboard
Manage Tools View Help
Dashboard
Local Server
All Servers
AD DS
DHCP
DNS
File and Storag
WELCOME TO SERVER MANAGER
Group Policy Management
File Action View Window Help
Group Policy Management
Forest: Meridian.local
Domains
Meridian.local
Default Domain Policy
Domain Password policy
Administration
ClientSupport
ClientSupport_PCs
DataService_PCs
DataServices
Domain Controllers
Governance
Governance_PCs
Groups
Group Policy Objects
WMI Filters
Starter GPOs
Sites
Group Policy Modeling
Group Policy Results
Meridian.local
Status
Linked Group Policy Objects
Group Policy Inheritance
Delegation
Link Order
GPO
Enforced
Link Enabled
1
Default Domain Policy
No
Yes
2
Domain Password p...
No
Yes
Hide
Right Ctrl


File Machine View Input Devices Help
Server Manager
Server Manager ‣ Dashboard
Manage Tools View Help
Dashboard
Local Server
All Servers
AD DS
DHCP
DNS
File and Storag
WELCOME TO SERVER MANAGER
Group Policy Management
File Action View Window Help
Group Policy Management
Forest: Meridian.local
Domains
Meridian.local
Default Domain Policy
Domain Password policy
Administration
ClientSupport
ClientSupport_PCs
DataService_PCs
DataServices
Data Service Policy
Username: dmolefe
Username: mjacobs
Domain Controllers
Governance
Governance_PCs
Groups
Group Policy Objects
WMI Filters
Starter GPOs
Sites
Group Policy Modeling
DataServices
Linked Group Policy Objects
Group Policy Inheritance
Delegation
Link Order
GPO
Enforced
Link Enabled
1
Data Service Policy
No
Yes
Hide
Right Ctrl


File Machine View Input Devices Help
Server Manager
Server Manager • Dashboard
Manage Tools View Help
Dashboard
Local Server
All Servers
AD DS
DHCP
DNS
File and Storag
WELCOME TO SERVER MANAGER
Group Policy Management
File Action View Window Help
Group Policy Management
Forest: Meridian.local
Domains
Meridian.local
Default Domain Policy
Domain Password policy
Administration
ClientSupport
ClientSupport_PCs
DataService_PCs
DataServices
Domain Controllers
Governance
Governance_PCs
Groups
Group Policy Objects
WMI Filters
Starter GPOs
Sites
Group Policy Modeling
Group Policy Results
Governance
Linked Group Policy Objects
Group Policy Inheritance
Delegation
Link Order
GPO
Enforced
Link Enabled
1
Governance USB R...
No
Yes
Hide
Right Ctrl

3.1b)


File Machine View Input Devices Help
Server Manager • Dashboard
Manage Tools View Help
Group Policy Management Editor
File Action View Help
All Removable Storage classes: Deny all access
All Removable Storage classes: Deny all access
Previous Setting
Next Setting
Not Configured
Enabled
Disabled
Comment:
Supported on:
At least Windows Vista
Options:
Help:
Configure access to all removable storage classes.
This policy setting takes precedence over any individual removable storage policy settings. To manage individual classes, use the policy settings available for each class.
If you enable this policy setting, no access is allowed to any removable storage class.
If you disable or do not configure this policy setting, write and read accesses are allowed to all removable storage classes.
19 setting(s)
Hide
Right Ctrl

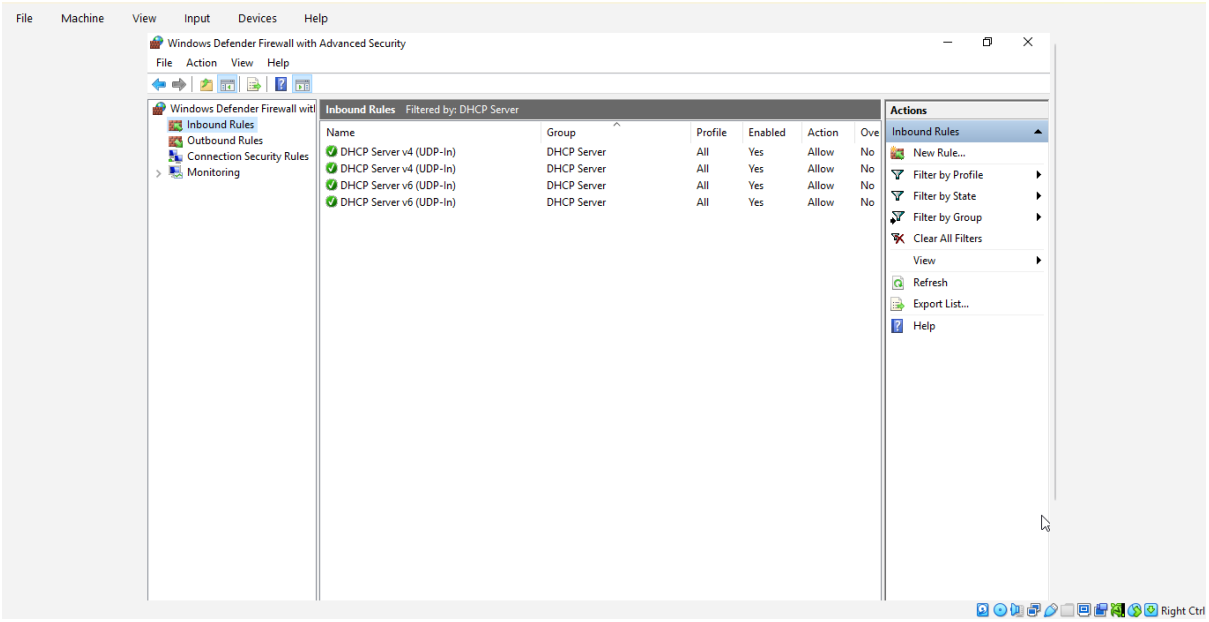
With the Data services policy, after following the step I could not find the way that could allow me to disable the command prompt thus assuming the machine took care of the situation beforehand.

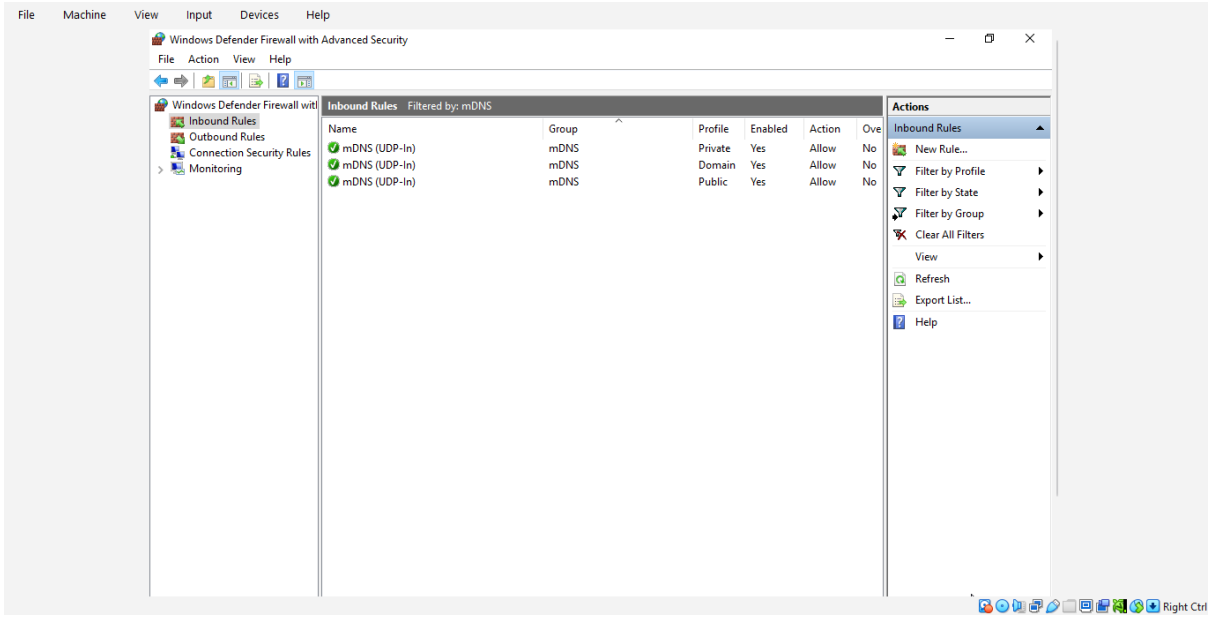
3.2d) The reason why the internal CA is more suitable than self-signed certificates is that an internal certification authority is suitable because it establishes trusts for domain joined devices automatically. This helps manage certificate renewal and issuance. Self-signed certificates generate browser warning and are difficult to manage across multiple systems.

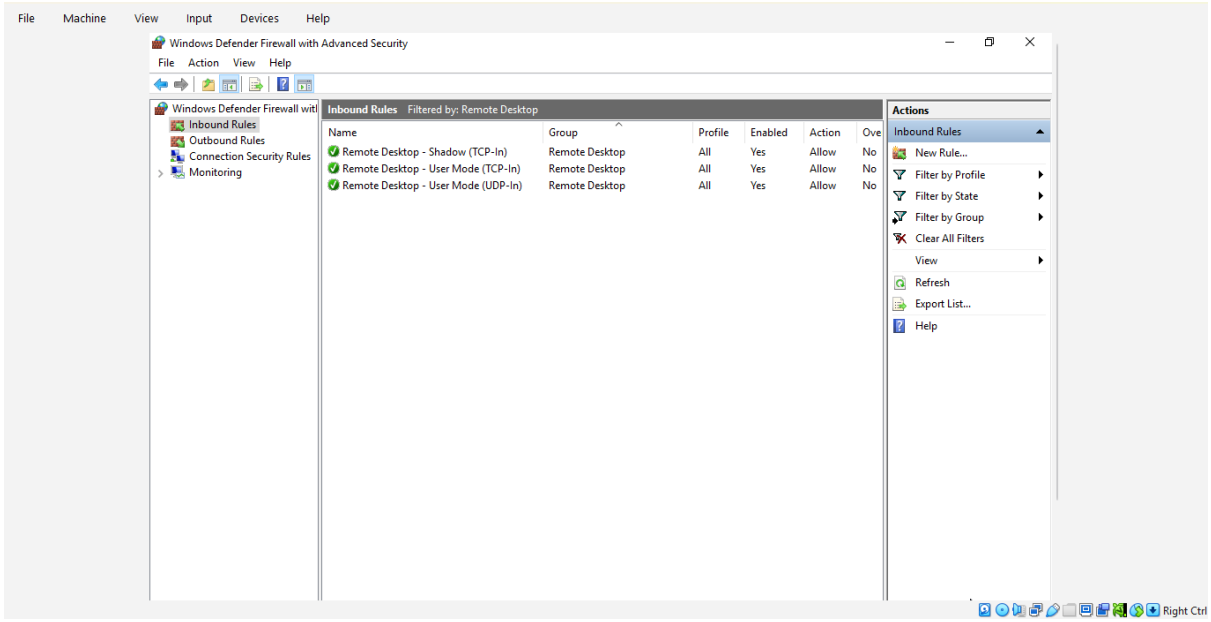
HTTPS is necessary for the internal portal because it encrypts communication between users and the portal. This protects sensitive information from being intercepted and verifies the identity of the website through a trusted certificate.

DELIVERABLE 4

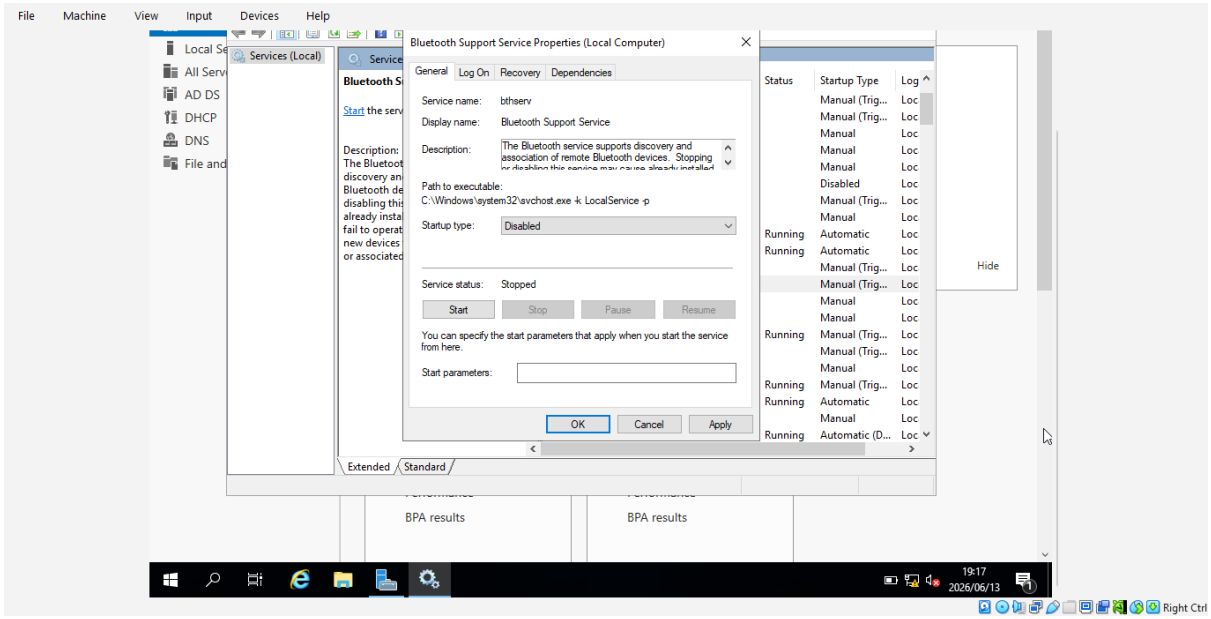
4a)

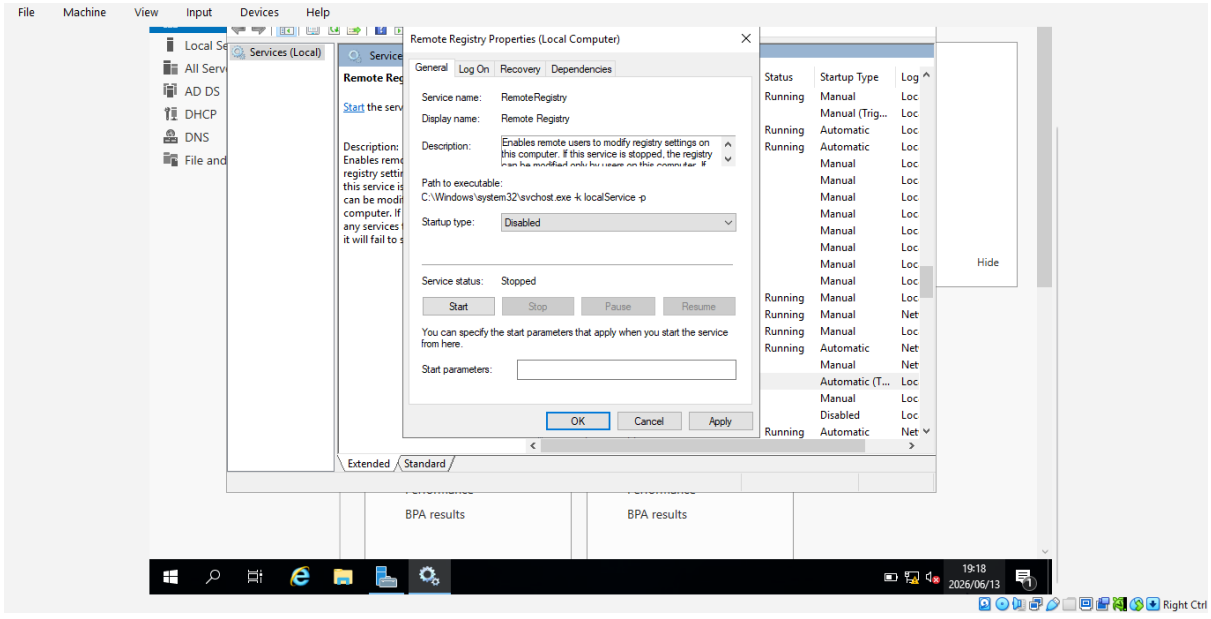


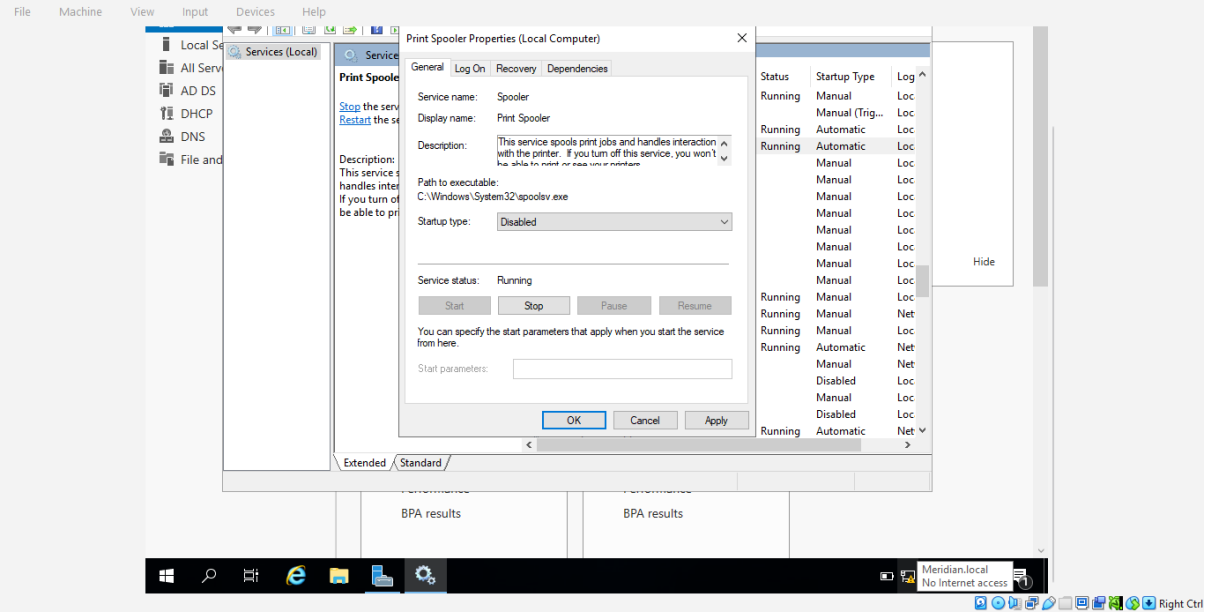

File Machine View Input Devices Help
Windows Defender Firewall with Advanced Security
File Action View Help
Windows Defender Firewall with
Inbound Rules
Outbound Rules
Connection Security Rules
Monitoring
Inbound Rules Filtered by: mDNS
Name Group Profile Enabled Action Ove
mDNS (UDP-In) mDNS Private Yes Allow No
mDNS (UDP-In) mDNS Domain Yes Allow No
mDNS (UDP-In) mDNS Public Yes Allow No
Actions
Inbound Rules
New Rule...
Filter by Profile
Filter by State
Filter by Group
Clear All Filters
View
Refresh
Export List...
Help
Right Ctrl


File Machine View Input Devices Help
Windows Defender Firewall with Advanced Security
File Action View Help
Windows Defender Firewall with
Inbound Rules
Outbound Rules
Connection Security Rules
Monitoring
Inbound Rules Filtered by: Remote Desktop
Name Group Profile Enabled Action Ove
Remote Desktop - Shadow (TCP-In) Remote Desktop All Yes Allow No
Remote Desktop - User Mode (TCP-In) Remote Desktop All Yes Allow No
Remote Desktop - User Mode (UDP-In) Remote Desktop All Yes Allow No
Actions
Inbound Rules
New Rule...
Filter by Profile
Filter by State
Filter by Group
Clear All Filters
View
Refresh
Export List...
Help
Right Ctrl

b)


File Machine View Input Devices Help
Bluetooth Support Service Properties (Local Computer)
General Log On Recovery Dependencies
Service name: bthserv
Display name: Bluetooth Support Service
Description: The Bluetooth service supports discovery and association of remote Bluetooth devices. Stopping
Path to executable:
C:\Windows\system32\svchost.exe -k LocalService -p
Startup type: Disabled
Service status: Stopped
Start Stop Pause Resume
You can specify the start parameters that apply when you start the service from here.
Start parameters:
OK Cancel Apply
Services (Local)
Extended Standard
BPA results
BPA results
Hide
19:17
2026/06/13
Right Ctrl


File Machine View Input Devices Help
Remote Registry Properties (Local Computer)
General Log On Recovery Dependencies
Service name: RemoteRegistry
Display name: Remote Registry
Description: Enables remote users to modify registry settings on this computer. If this service is stopped, the registry
Path to executable:
C:\Windows\system32\svchost.exe -k localService -p
Startup type: Disabled
Service status: Stopped
Start Stop Pause Resume
You can specify the start parameters that apply when you start the service from here.
Start parameters:
OK Cancel Apply
Services (Local)
Extended Standard
BPA results
BPA results
Hide
19:18
2026/06/13
Right Ctrl



d) The one dependency deployed between services is Active Directory Domain system. It depends on the DNS because domain joined computers use DNS to locate Domain Controllers and authenticate users. Without DNS, users cannot log into the domain.

The one limitation of the pilot environment is that it only uses one Domain Controller. If that server fails, DNS, Authentication and other domain services become unavailable.

The one production that could improve deployment is an additional Domain Controller. It provides redundancy and fault tolerance. This ensures that users can still authenticate if one server fails or becomes unavailable.

The one advantage of using group based administration is that it allows permissions and policies to be applied to groups instead of individual users. This will reduce administrative effort and improve consistency and makes security management easier

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